

Task 2 Report: K-Means Movie Recommender

Practical Course Algorithms for AI 3 (RJ) - Lab Work 3

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Introduction

This task studies a simple movie recommender engine using the K-Means algorithm. The data contains 10 users, and each user has watched 10 movies. Every movie is described by genre attributes, such as Comedy, Drama, Action, Thriller, and Crime. First, the watched movies of each user were converted into one user profile by adding the genre values. Then K-Means was used to group users with similar movie interests.

Goal

The goal is to find groups of similar users and then use one selected group for a simple recommendation. If one user has not watched a movie, but other users in the same cluster watched it, that movie can be suggested to the selected user.

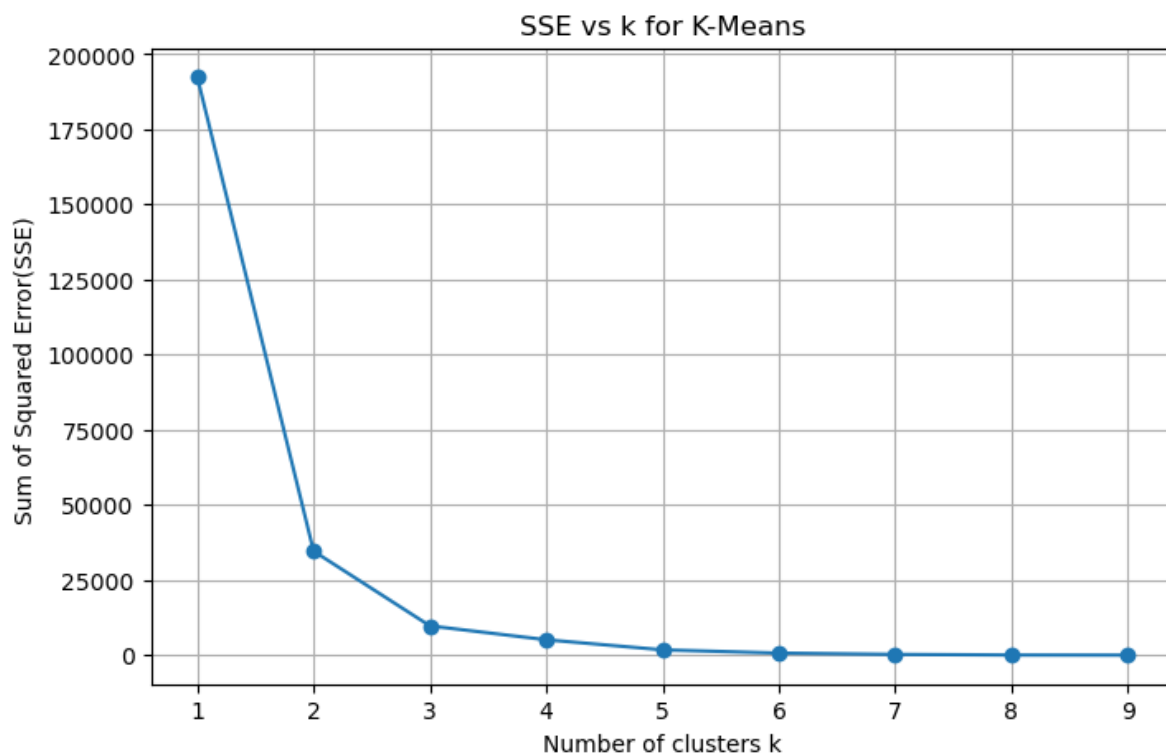
a) Generated source file

The ten user movie files were combined into one source file. For each user, the genre attributes were accumulated by summing the attribute values of the watched movies. This gives one numeric profile for each user, which is then used as the input for K-Means.

User	Romance	Comedy	Science	Fiction	Drama	Romantic	Fantasy	Animation	Adventure	Action	Thriller	Erotic	Crime	index	rating
u1	2	4	1	1	3	3	1	0	0	0	0	0	0	45	0
u2	1	6	0	0	5	4	0	1	1	0	0	0	0	205	2
u3	1	5	1	1	7	2	0	1	1	0	0	0	0	405	7
u4	0	6	0	0	7	3	0	2	1	0	1	0	1	449	5
u5	1	6	0	0	5	4	0	1	1	0	0	0	0	225	3
u6	2	6	0	0	5	1	0	0	0	1	1	0	0	280	4
u7	0	4	0	0	4	2	0	1	0	3	4	0	1	406	2
u8	0	5	2	2	5	1	1	1	2	2	1	1	0	451	4
u9	1	7	0	0	3	8	1	0	0	0	0	0	0	175	2
u10	0	4	2	2	4	1	1	1	1	4	2	0	0	475	5

b) SSE plot and c) selected k value

K-Means was tested for different values of k. The Sum of Squared Error (SSE) was plotted against k to find the elbow point.



The SSE falls strongly from $k = 1$ to $k = 3$. After $k = 3$, the curve becomes much flatter and the improvement is smaller. Therefore, $k = 3$ was selected as a suitable number of clusters.

d) Final K-Means clusters for $k = 3$

Cluster	Users in the cluster
Cluster 0	u3, u4, u7, u8, u10
Cluster 1	u1
Cluster 2	u2, u5, u6, u9

e) Simple recommendation process by hand

For the manual recommendation step, Cluster 0 was selected because it contains several users. User u7 was chosen as the target user. The idea is simple: users in the same cluster are treated as similar, so movies watched by other users in the same cluster but not watched by u7 can be recommended to u7.

Steps	Result
Selected cluster	Cluster 0
Users in this cluster	u3, u4, u7, u8, u10
Selected target user	u7
Movies already watched by u7	Ratatouille; Skyfall; Marley and Me; The Dark Knight; Midnight in Paris; No Country for Old Men; My Week with Marilyn; I Care a Lot; Spider-Man 2; Casablanca
Recommended movies for u7	Everything Everywhere All at Once; Iron Man 3; Alice in Wonderland; Jane Eyre; X-Men; The Fallout; Avatar; Anora; Toy Story; Quantum of Solace; Not Easily Broken; A Serious Man; The Contractor; Superman Returns; Pirates of the Caribbean; Across the Universe; Spring Breakers; Fixed; Infinite; Beginners; Babygirl; New Years Eve.

Short comment on the result: The recommendation is reasonable for this small experiment because it only uses users from the same K-Means cluster. However, the result depends strongly on the selected cluster and on the small number of users. With more users and real ratings, the recommendation would probably become more reliable.